

Haptic Devices with Variable Volume Using Spiral Springs

Minhyeok Baek

minheak06@dgist.ac.kr

Daegu Gyeongbuk Institute of Science and Technology

Daegu, Republic of Korea

Sunjun Kim*

sunjun_kim@dgist.ac.kr

Daegu Gyeongbuk Institute of Science and Technology

Daegu, Republic of Korea

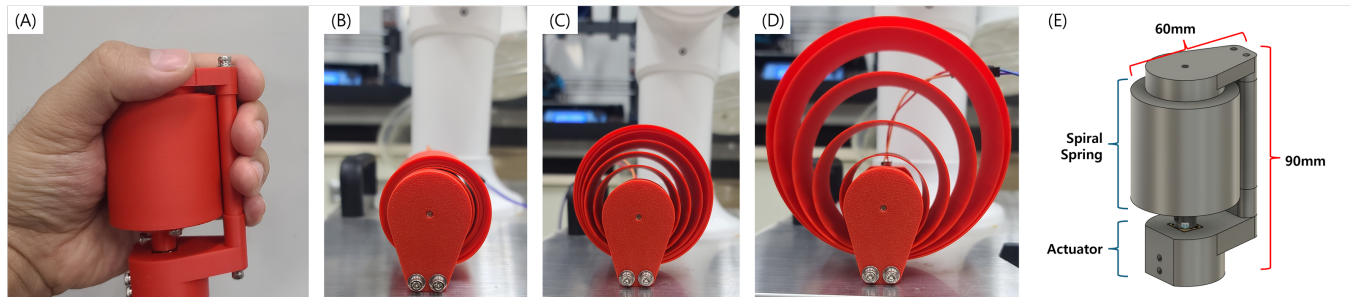


Figure 1: A. The appearance of the haptic device being worn B-D. Deformation area of the spiral spring, with B representing the minimum size and D representing the maximum size. E. 3D modeling of Spiral spring haptic device and simple description.

ABSTRACT

Controllers that convey tactile sensations of object size in virtual environments are crucial for user interaction. However, existing research often faces commercialization constraints due to the complexity and high number of actuators required. This study proposes a haptic device that utilizes a spiral spring structure that offers approximately three times the variable range with a single actuator. This system can quickly render the diameters of objects grasped by users and simulate continuous volume changes, such as inflating a balloon. Due to its simple structure and wide variable range, this system is expected to be suitable for various scenarios, providing users with a more immersive virtual reality experience.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

Virtual reality, Handheld haptics, Shape-changing, Spiral spring, Grasping

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*Correspondence author

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1 INTRODUCTION

Rendering the tactile sensation of objects that users touch in virtual environments is a very important task. When there is a discrepancy between the physical tactile sensation felt when manipulating various objects in a virtual environment and their representation in virtual reality, it can diminish the user's sense of immersion. To address this, extensive research has focused on haptic controllers that render tactile feedback for virtual objects using various methods. Studies have been conducted on using multiple actuators, such as Wolverine and CyberGrasp, to provide physical resistance to each finger to recognize shapes [2]. Additionally, in order to convey more abstract and diverse physical sensations, various shapes have been implemented using pneumatic actuators like Pupop [6], or actuators that can protrude at the contact points, as seen in devices like X-Rings and PoCoPo [3][7]. Other approaches have moved away from the traditional controller form, proposing designs such as Haptobend, which can freely deform into various physical shapes [5], or Shape-A-Getti, which modifies the entire grip structure through rotation [1]. Recent studies have focused on devices that induce haptic illusions by altering the position of the fingers without changing the physical shape [4].

While these attempts enhance the user's sensory experience, they have the downside of being complex with many actuators. Even with pneumatic systems instead of motors, additional devices are required, and these often have a limited range of volume change relative to their size and structure. This limitation becomes more pronounced when interacting with objects of large diameters. Such complex devices and structures present significant challenges for commercialization to actual users.

Therefore, this study proposes a new haptic structure that can be produced using only a single motor and has a very large variable range. The key idea is to use a spiral spring structure, allowing for a very large volume change with a single motor. This approach enables the simulation of various physical shapes with a simple structure.

The proposed spiral spring haptic device can deliver immersive haptic sensations, enhancing interaction in virtual reality.

2 IMPLEMENTATION

2.1 Hardware Overview

The controller comprises a simple actuator and a spiral spring, as depicted in Figure 1A and 1E. An Arduino Pro Micro from SparkFun is employed to control the actuator and facilitate communication with the computer. All components were fabricated using PLA through 3D printing. The device dimensions are approximately $60 \times 100 \times 50$ mm.

2.2 Variable Structure

The controller employs a spiral spring mechanism. As shown in Figure 1 B-D, the volume is modulated by rotating the center of the spring structure to wind or unwind the spring. Users can grasp the outer wall of the spring and perceive the volumetric changes as the structure alters.

Its straightforward construction and compact size enable comfortable use in one hand, as illustrated in Figure 1. It has a diameter of 45 mm at maximum compression and 120 mm at maximum expansion, nearly tripling its compressed width.

3 DEMO SCENARIO

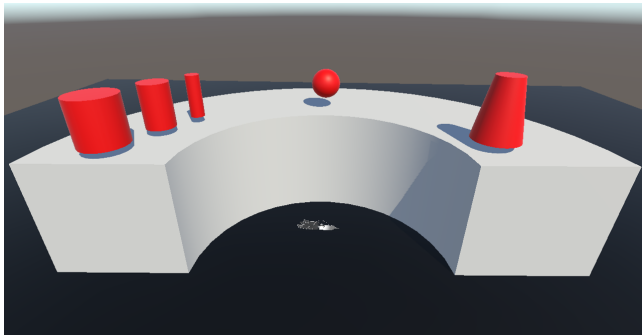


Figure 2: Implemented virtual environments: from left to right, the cylinder, balloon, and frustum scenarios.

Three scenarios demonstrate the advantages of the spiral spring structure. These scenarios were developed in Unity and implemented in a virtual environment using the Quest 3 by Meta, as illustrated in Figure 2.

3.1 Multiple objects of varying diameters (Cylinder scenario)

The user can interact with three cylinders of different diameters (with a diameter ratio of 1:2:3). They can freely manipulate the cylinders with their hands, and when grasping each cylinder, the spiral spring compresses or expands to match the corresponding diameter, providing the user with a realistic haptic sensation.

3.2 Objects with continuously varying diameters (Balloon scenario)

The user can interact with a sphere that changes in volume. As the volume of the sphere continuously varies, the user can adjust its diameter using a controller. As the user modifies the sphere's diameter, the controller changes its volume through the spring's expansion and contraction, providing corresponding tactile feedback.

3.3 Single objects with varying diameters (Frustum scenario)

The user can interact with an object, such as a frustum of a cone, whose diameter changes based on their position. The diameter varies depending on where the user touches the object, allowing them to experience a continuously changing diameter as they move their touch along the surface.

4 CONCLUSION

In this study, we propose a spiral spring haptic structure with a significantly large variable area using only one actuator. This structure excels in its high variable ratio and ability to simulate continuous changes, offering a unique virtual reality experience compared to other haptic devices.

5 FUTURE PLANS

5.1 Physical analysis of spiral springs

The current version utilizes a custom-made spiral spring. One of the primary advantages of this study is the simplicity of the structure, which facilitates easy replacement of individual components. By conducting a physical analysis of the spiral spring, we can examine the modeling factors (such as spring thickness, total diameter, and spring spacing) and the variable volume. This analysis enables the creation of a spring tailored to specific scenarios. Consequently, this approach allows for the development of a more optimized and modular haptic controller.

5.2 Hardware improvements

The current haptic controller only simulates a spiral spring. Future developments will integrate basic interaction elements, such as buttons, creating a haptic controller with functionalities similar to commercial controllers. This advancement will enable the development of a controller capable of rendering volumetric changes while preserving the size and portability of traditional controllers.

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